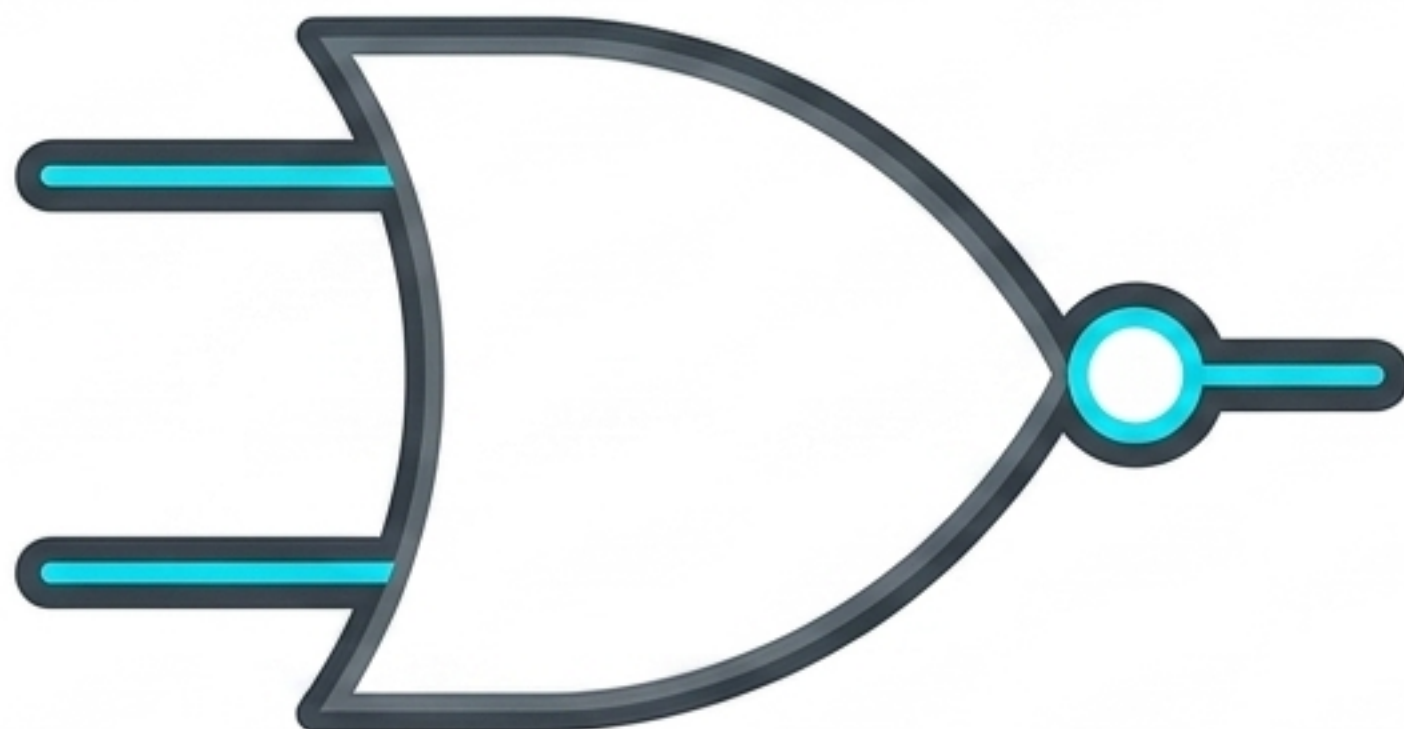




Universal Logic Generation

Step 1: The Core Concept

NAND and NOR are **Universal** gates. They act as the ultimate foundation for digital architecture.

Step 2: The Power of Universality

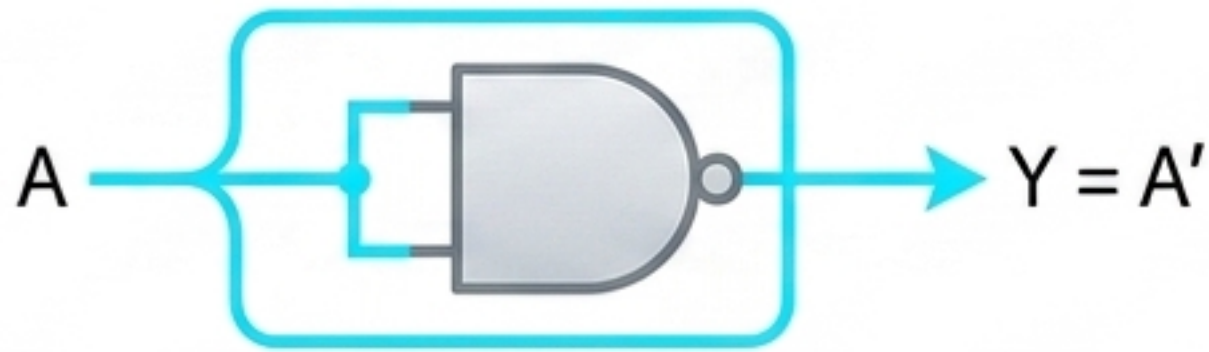
These gates can implement absolutely **ANY** basic Boolean function. No other gate types are required.

Step 3: Maximum Efficiency

Using 1 single gate type drastically simplifies hardware manufacturing and reduces transistor count.

Generating NOT and AND

1



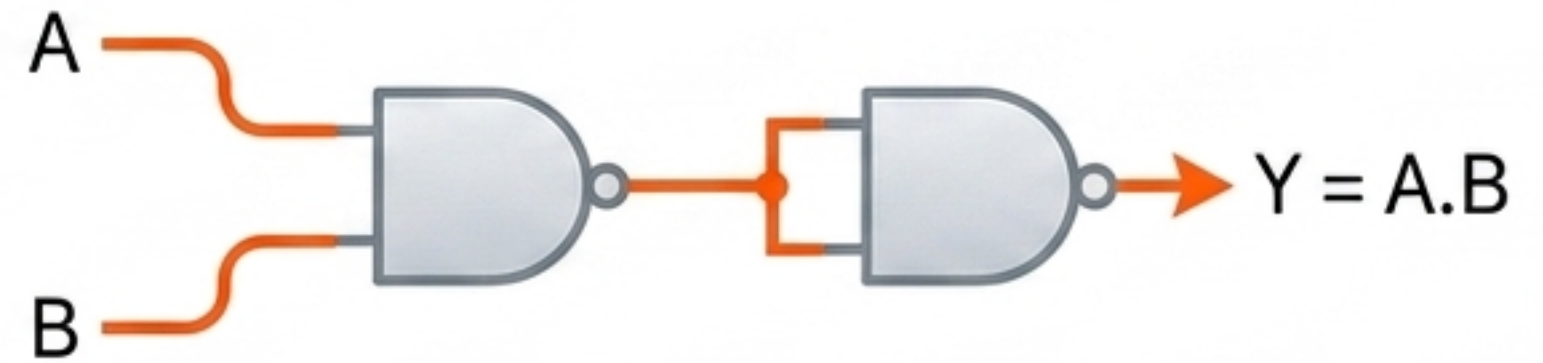
Panel 1: NOT Gate

Step 1: Connect 1 universal gate.

Step 2: Short the inputs together.

Result: The gate simply inverts the signal. (Input A becomes Output A').

2



Panel 2: AND Gate

Step 1: Connect 2 universal gates in series.

Step 2: The 1st gate computes the NAND function.

Step 3: The 2nd gate (shorted) inverts that output, perfectly restoring the AND logic.

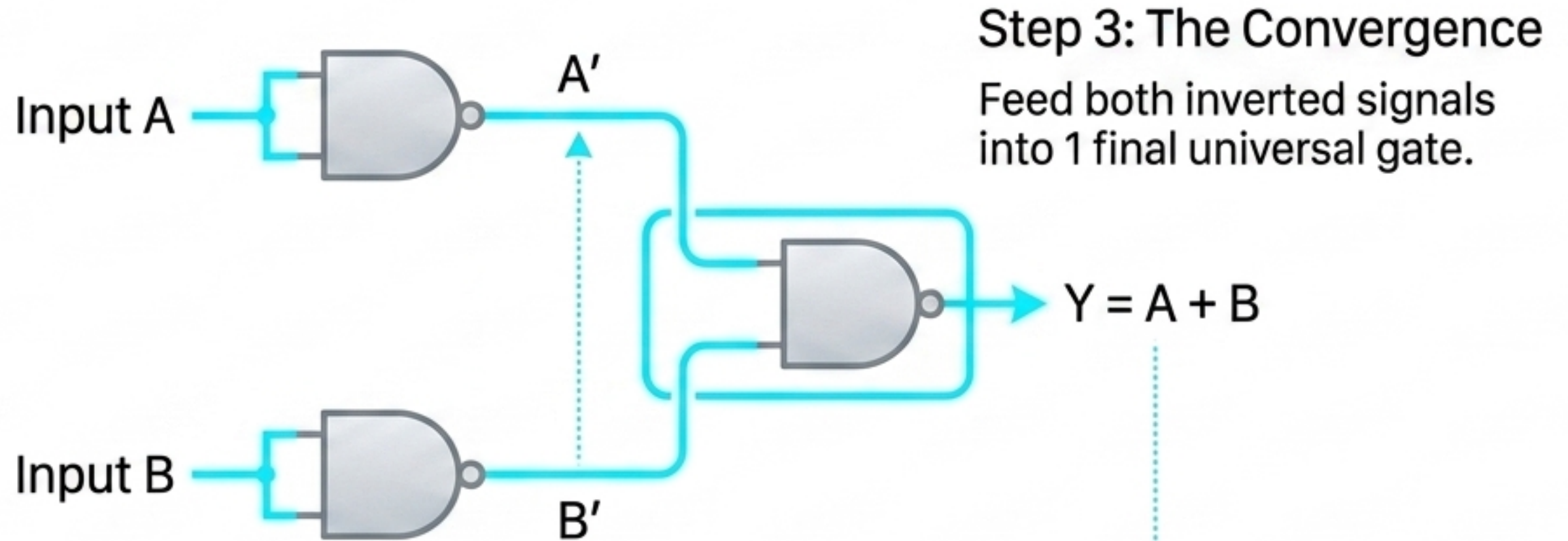
Generating OR Gates

Step 1: The Configuration

Connect 3 universal gates together to form an OR function.

Step 2: The Input Stage

Use 2 gates to individually invert input A and input B.



Callout Box: DeMorgan's Theorem

Inverting the inputs before combining them flips the logic from AND to OR.

$$(A' * B')' = A + B$$

The Universal Multi-Tool

1 Step 1: The Concept
1 tool containing multiple functions.

2 Step 2: Connect 1 Gate
Acts as Scissors.
Digital equivalent: NOT function
(Simple inversion and cutting).

3 Step 3: Connect 2 Gates
Acts as a Knife.
Digital equivalent: AND function
(Precise, linear combination).

4 Step 4: Connect 3 Gates
Acts as a Screwdriver.
Digital equivalent: OR function
(Broad, multi-angle application).

